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Blood-based circulating microRNAs as diagnostic biomarkers in cutaneous melanoma: a systematic review and meta-analysis

Hendra Pramono[1,4], Ana L. Ekowati[2,4], Samuel H. Pinandyo[1] & Ferbian M. Siswanto[3]✉

With the ever-increasing incidence of cutaneous melanoma (CM), early detection has become pivotal for ensuring patient survival. MicroRNAs (miRNAs) are promising diagnostic biomarkers owing to their stability and wide availability in various body fluids. This systematic review and meta-analysis evaluated the diagnostic performance of blood-based circulating free miRNAs and extracellular vesicle (EV)-derived miRNAs for CM. The study protocol was registered in PROSPERO (CRD420251125262). A total of 557 records were identified, with 11 studies included in the systematic review and 10 in the meta-analysis. Diagnostic performance was evaluated for all studies combined, and separately for individual miRNA and miRNA combinations. The bivariate model showed a pooled sensitivity of 0.86, specificity of 0.85, and diagnostic odds ratio (DOR) of 33.52. Separate analyses indicated that individual miRNAs yielded sensitivity, specificity, and DOR values of 0.80, 0.78, and 13.71, respectively. On the other hand, miRNA combinations achieved superior accuracy with a sensitivity of 0.91, specificity of 0.89, and DOR of 84.05. Subgroup analysis demonstrated that the plasma samples outperformed the other sample sources. Overall, circulating miRNAs exhibit strong potential as noninvasive diagnostic biomarkers for CM.

Cutaneous melanoma (CM) is an aggressive form of skin neoplasia that originates from melanocytes. The incidence of CM has been increasing worldwide and is predicted to continue rising. In 2020, GLOBOCAN estimated approximately 325,000 new cases of CM and 57,000 related deaths. By 2040, the incidence is expected to increase by more than 50%, while the mortality rate may increase to 68%[1]. In highly metastatic diseases such as CM, early diagnosis plays a crucial role in improving patient survival. Current diagnostic methods still rely heavily on clinical assessment via dermoscopy or imaging techniques, whereas a definitive diagnosis requires histopathological examination of a lesion biopsy[2]. However, histological examinations are limited by subjectivity and inconsistency in evaluation. Therefore, the detection of reliable and accessible biomarkers for the early diagnosis of CM has become essential to reduce the reliance on expert interpretation and the availability of specialized diagnostic tools[3].

MicroRNAs (miRNAs) are small noncoding RNAs consisting of 17–25 nucleotides[4]. The ability of miRNAs to regulate multiple mRNAs is mediated through interactions with various regions in mRNAs, such as 3' untranslated regions (3' UTRs), 5' untranslated regions (5' UTRs), coding sequences (CDSs), and gene promoters. Alterations in miRNA expression play a role in various cellular functions and may contribute to the development of numerous diseases, including CM. Although malignant transformation is primarily caused by inadequate DNA repair against ultraviolet radiation exposure[5], both upregulation and downregulation of miRNAs have been described to play several roles in CM progression, such as proliferation[6], invasion[7], and metastasis[8], as these aberrations reflect DNA damage and ultraviolet-induced inflammatory responses[9].

Numerous studies have shown that miRNAs hold great potential as biomarkers since miRNAs can be found circulating in various body fluids. They are secreted into the extracellular space either within vesicles (microvesicles, ectosomes, and exosomes) or bound to proteins such as Argonaute 2 (AGO2), HDL, and Nucleophosmin 1 (NPM1). Through this mechanism, miRNAs remain stable under various conditions despite

[1]Bachelor of Medicine Study Program, School of Medicine and Health Sciences, Atma Jaya Catholic University of Indonesia, Jakarta, Indonesia. [2]Department of Medical Biology, School of Medicine and Health Sciences, Atma Jaya Catholic University of Indonesia, Jakarta, Indonesia. [3]Department of Chemistry and Biochemistry, School of Medicine and Health Sciences, Atma Jaya Catholic University of Indonesia, Jakarta, Indonesia. [4]These authors contributed equally to this work: Hendra Pramono and Ana L. Ekowati. ✉email: ferbian.siswanto@atmajaya.ac.id

the presence of endogenous ribonucleases, extreme temperatures, pH variations, and freeze–thaw cycles. Once released, the miRNA-containing structures enter into biofluids such as blood and saliva[10,11]. In CM patients, several miRNAs have been reported to be dysregulated. The difference in miRNA expression can subsequently affect melanoma specific target genes, thereby driving oncogenic processes[12].

Although a small proportion of circulating miRNAs are encapsulated within vesicles, several studies have indicated that extracellular vesicle (EV)-miRNAs outperform circulating-free miRNAs in terms of diagnostic performance[13,14]. EV-miRNAs display a distinct compositional profile due to possible selective sorting mechanisms that result in the enrichment of cancer-specific miRNA signatures[15]. However, to date, no systematic review and meta-analysis has specifically addressed blood-based circulating miRNAs and EV-miRNAs in CM. Therefore, here we performed a systematic review and meta-analysis of diagnostic test accuracy incorporating the recent findings on both blood-based circulating-free miRNAs and EV-miRNAs in CM patients compared with controls.

Results

Study selection

A total of 535 studies were identified from the PubMed, ProQuest, and EBSCO databases. After 12 duplicates and nine retracted studies were removed, titles and abstracts of the remaining 514 studies were screened. Among the 26 studies evaluated through full-text screening, 10 studies were eligible[16–25]. Additionally, 22 studies were identified through citation searching, with one study ultimately included after full-text screening[26]. Among the excluded studies, one reported the diagnostic performance of EV-miRNAs. A Study by Tengda et al.[27] was excluded due to the inclusion of patients with melanoma types other than CM. A Study by Sabato et al.[25] was included in the systematic review but excluded from meta-analysis because of insufficient data. The workflow is presented in the Preferred Reporting Items for a Systematic Review and Meta-analysis (PRISMA) diagram in Fig. 1.

Study characteristics

Ten studies comprising 1,154 patients and 691 controls were included in the meta-analysis. Three studies reported results for both individual miRNAs and combinations of multiple miRNAs. Two studies employed detection methods other than qRT–PCR, and only the study by De Martino et al.[16] evaluated extracellular vesicle (EV)-miRNAs as biological samples. The characteristics of each study are provided in Supplementary Table S1, while the biological relevance of each miRNAs obtained from each study is presented in Table 1.

Risk of bias and applicability

Based on the evaluation using Quality Assessment of Diagnostic Accuracy Studies-2 (QUADAS-2) tool, the included studies showed an overall high risk of bias in the patient selection domain since most of the studies were performed with a case-control design. Seven studies were considered to have a high risk of bias in the index test domain because there was no prespecified threshold, whereas four other studies were unclear because of insufficient information. Overall, studies were considered to have a low risk of bias in the reference standard domain and flow and timing domain. All included studies displayed no concerns regarding applicability. The assessment of the included studies is provided in Fig. 2.

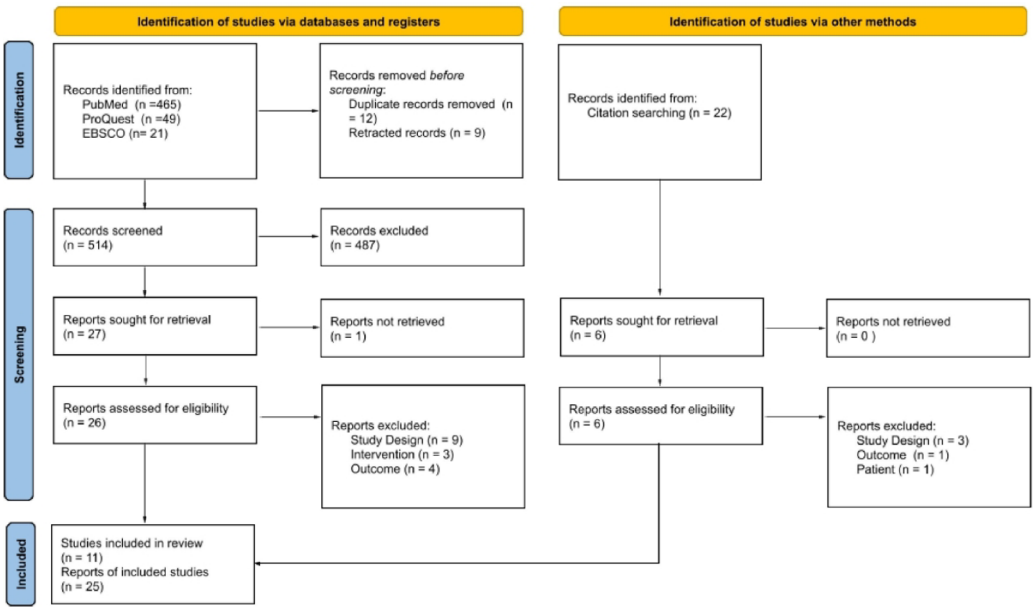


Fig. 1. PRISMA diagram showing the research workflow[37]. A total of 557 studies were screened, with 11 studies eligible for the systematic review and 10 studies included in the meta-analysis, consisting of 25 reports.

miRNAs	Expression	Matrices	Effect	References
hsa-miR-200c-3p	Up	Plasma exosome	Inhibits EMT*	38,39
hsa-miR-144-3p	Down	Plasma exosome	Inhibits proliferation and invasion; correlated	40,41
			with melanoma ulceration	
hsa-miR-221-3p	Down	Plasma exosome	Overexpression associated with p27-mediated	42,43
			tumor proliferation; drives melanoma develop-	
			ment and dissemination	
miR-1246	Up	Plasma	Promotes cell proliferation and metastasis by	8
			suppressing Bax expression, promoting Bcl-2,	
			and targeting FOXA2	
miR-185	Up	Plasma	Inhibits tumor growth by silencing IL10Rα; in-	44,45
			hibits proliferation and migration	
miR-20a	Up	Plasma	Increases cell growth, migration, and invasion*	46
let-7b	Up	Plasma	Tumor suppressor; inhibits cell cycle progres-	47
			sion	
miR-338-3p	Up	Plasma	Inhibits proliferation, migration, and invasion	48
miR-149-3p	Up	Plasma	Induces apoptosis resistance through binding	49
			with Mcl-1	
miR-150-5p	Up	Plasma	Predicts brain metastasis; inhibits proliferation,	48
			migration, and invasion of melanoma cells	
miR-193a-3p	Down	Plasma	Inhibits cell proliferation	6
miR-15b-5p	Up	Plasma	Associated with poor prognosis and brain metas-	48,50
			tasis	
miR-524-5p	Down	Plasma	Inhibits proliferation and migration by suppress-	48
			ing the MAPK/ERK signaling pathway	
miR-16	Down	Serum	Inhibits proliferation and metastasis	35
miR-320a-3p	Down	Plasma	Promotes susceptibility to cytotoxic T lympho-	51
			cytes	
miR-134-5p	Down	Plasma	Inhibits proliferation, migration, invasion, and	48
			induces cell cycle arrest and apoptosis by sup-	
			pressing CTHRC1	
miR-489	Down	Serum	Inhibits proliferation, migration, and invasion of	48
			melanoma cells	
miR-21	Up	Serum	Enhances tumor invasion; associated with	48
			chemosensitivity and radiosensitivity	
miR-10b	Up	Serum	Promotes independent growth and invasion in	7
			BRAF V600E melanoma cells	
miR-412-3p	Up	Blood	Promotes EMT*	52
miR-507	Up	Blood	Promotes cell proliferation and invasion by sup-	53
			pressing UCA1	
miR-1203	Up	Blood	Promotes invasion and metastasis by suppressing	54
			SOCS3*	
miR-362-3p	Down	Blood	Correlated with acral melanoma	30

Table 1. Circulating miRNAs and their biological relevance in melanoma. Notes: [*] Indicates biological effects inferred from studies in other malignancies. Abbreviations: EMT: epithelial–mesenchymal transition.

Diagnostic accuracy analyses

The sensitivity, specificity, and DOR of 11 independent studies were pooled. An analysis of all studies showed circulating miRNAs were able to achieve 0.860 (95% CI 0.792–0.909) in sensitivity, 0.849 (95% CI 0.762–0.908) in specificity, and 41.207 (95% CI 17.638–96.270) in DOR. The pooled sensitivity, specificity, and DOR of all studies are displayed in Fig. 3. Further analyses were performed separately for studies reporting individual miRNAs and miRNA combinations. Seven reports of individual miRNAs reported sensitivity, specificity, and DOR values of 0.806 (95% CI 0.736–0.860), 0.771 (95% CI 0.685—0.840), and 14.0 (95% CI 9.464–20.709) respectively, while six reports of miRNA combinations displayed a sensitivity of 0.915 (95% CI 0.857–0.950), specificity of 0.890 (95% CI 0.791—0.945), and DOR of 97.401 (95% CI 31.350–302.612). All studies resulted in substantial heterogeneity with I^2 ranging from 77.2% to 84.7% and $P < 0.0001$. Forest plots of the pooled sensitivity, specificity, and DOR of the circulating miRNAs are shown in Figs. 4 and 5. Subgroup analyses were conducted based on sample types (Fig. 3). Studies using plasma samples have shown higher values of sensitivity,

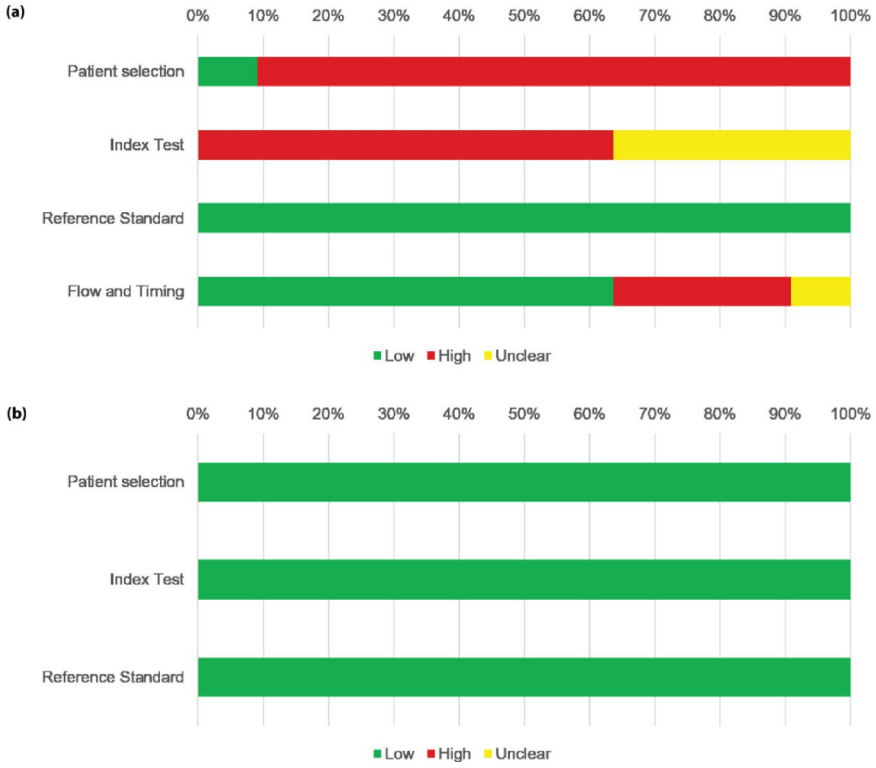


Fig. 2. Bar plot showing the results of the QUADAS-2 assessment of **(a)** risk of bias and **(b)** applicability.

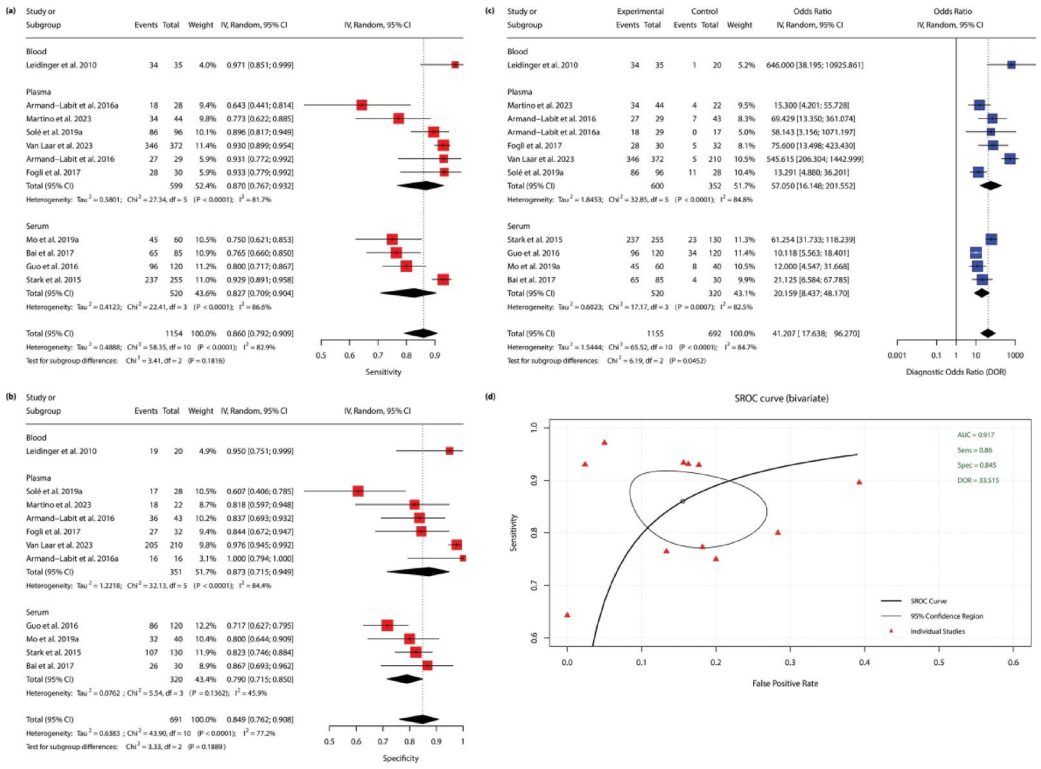


Fig. 3. Overall diagnostic performance of all studies. **(a)** Forest plot of sensitivity (sens), **(b)** specificity (spec), **(c)** diagnostic odds ratio (DOR), and **(d)** summary receiver operating characteristic (SROC) curve.

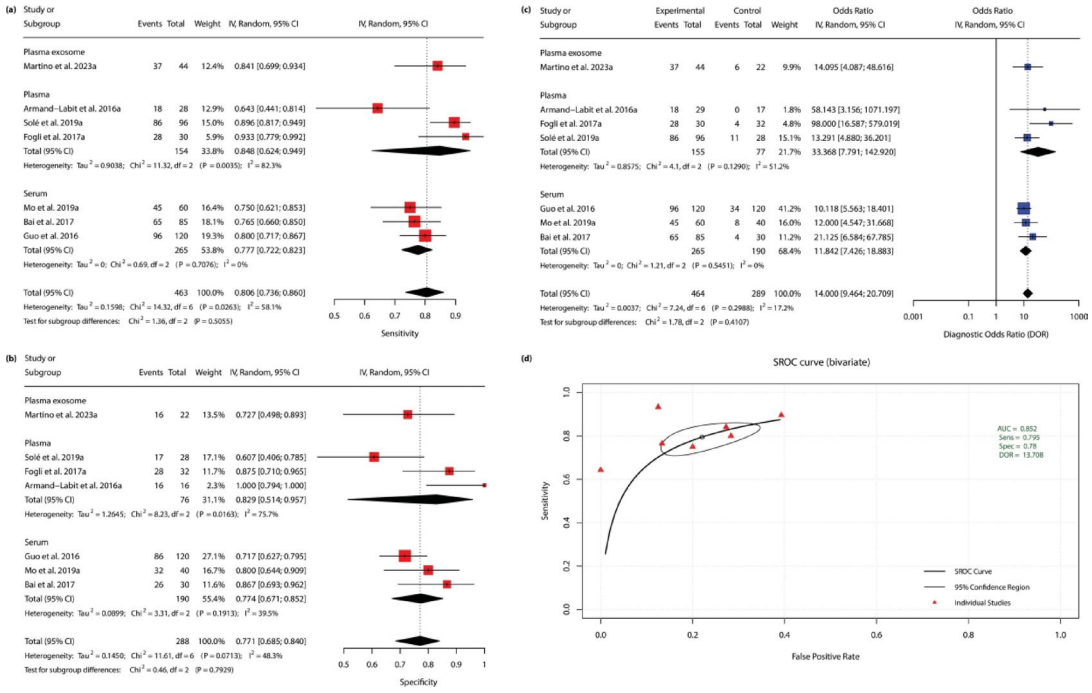


Fig. 4. Diagnostic performance of studies reporting individual miRNAs. **(a)** Forest plot of sensitivity (sens), **(b)** specificity (spec), **(c)** diagnostic odds ratio (DOR), and **(d)** summary receiver operating characteristic (SROC) curve.

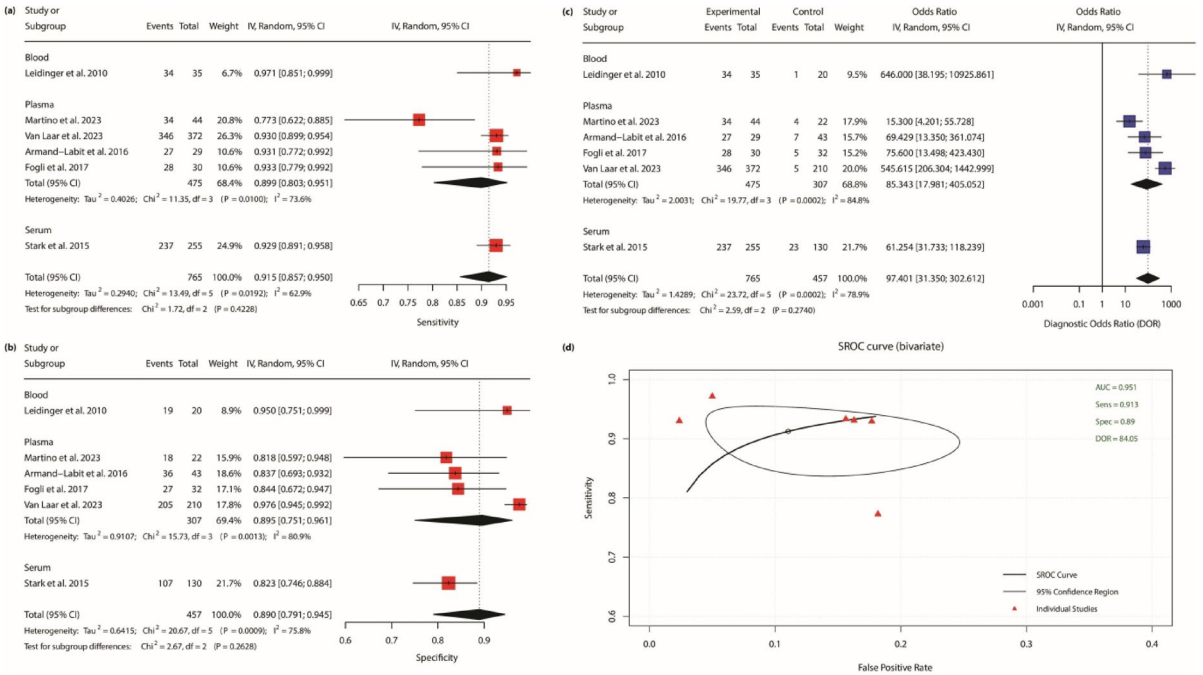


Fig. 5. Diagnostic performance of studies reporting miRNA combinations. **(a)** Forest plot of sensitivity (sens), **(b)** specificity (spec), **(c)** diagnostic odds ratio (DOR), and **(d)** summary receiver operating characteristic (SROC) curve.

specificity, and DOR compared to the other groups. A test for subgroup differences revealed that there were no significant differences in sensitivity or specificity ($P > 0.05$). However, significant differences were observed in DOR ($P = 0.0452$), indicating better performance of the plasma samples. A bivariate model was used to summarize the sensitivity, specificity, and DOR. The summary receiver operating characteristic (SROC) curves

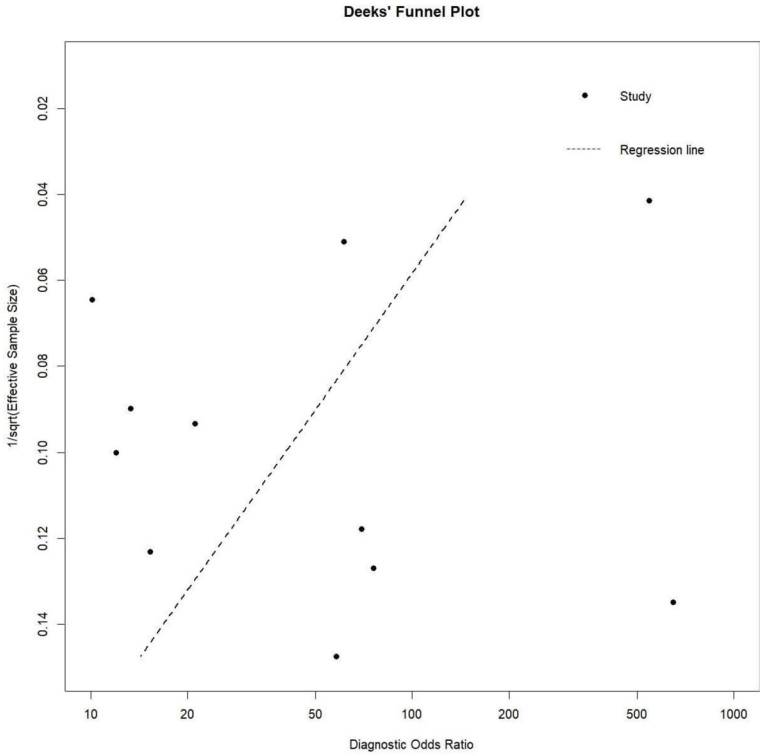


Fig. 6. Deeks' funnel plot of the included studies showing 11 reports of 10 studies.

were used to summarize the overall performance of all the studies as presented in Figs. 3D and 4D, and 5D. The overall performance of all studies was shown to have an area under the curve (AUC) value of 0.917, whereas the individual miRNAs and miRNA combinations had AUC values of 0.852 and 0.951, respectively. The correlation (ρ) between logit sensitivity and specificity for all studies was −0.347, suggesting that there was a trade-off between sensitivity and specificity.

Publication bias

Deeks' funnel plot asymmetry test on 10 included studies resulted in a t value of −1.496 and a P-value of 0.173, indicating that there was no publication bias. Deeks' funnel plot is displayed in Fig. 6.

Meta-regression

Bivariate meta-regression was performed to explore sources of heterogeneity in all studies. Sample type and type of miRNA reported (individual or combination) were used as covariates. The analysis revealed that the type of miRNA significantly affected sensitivity ($P = 0.004$), indicating that variation in reported miRNAs contributed to heterogeneity in diagnostic performance, with higher sensitivity for miRNA combinations. Analysis using other covariates did not reveal a significant effect on the sensitivity or false positive rate (FPR).

Discussion

The high mortality and increasing incidence of CM emphasize the necessity of more reliable diagnostic methods. Circulating miRNAs have been recognized as potential CM biomarkers crucial for early diagnosis. This systematic review included 11 studies of blood-based circulating miRNAs and reported a total of 83 miRNAs as diagnostic biomarkers for CM. Among those, 10 studies were included in the meta-analysis. Through a bivariate random-effects model, we found that blood-based circulating miRNAs exhibited excellent diagnostic performance for CM detection with an AUC greater than 0.9. To our knowledge this is the first systematic review and meta-analysis that reviewed the diagnostic performance of blood-based circulating miRNAs specifically for CM.

All reports of the included studies showed different miRNAs across samples, except for miR-16, which was reported in two different studies. This variation might be due to different sample types. Studies have shown that different sample types contain different amounts and types of miRNAs. During the coagulation process, miRNA molecules from blood cells are released, which can increase the concentration of some specific miRNAs in the serum. However, both plasma and serum have distinct advantages. Plasma is considered a better biomarker sample because of its ability to better represent truly circulating miRNAs. Moreover, serum miRNAs appear to be relatively stable as their composition does not significantly change over time after blood collection[28,29].

EVs are known to carry cancer-specific miRNAs. In this systematic review, several miRNAs that were previously associated with CM, such as miR-362-3p[25], miR-200c-3p[16], and miR-221, were found in EV-miRNAs[16]. Studies have reported that miR-362-3p is significantly upregulated in whole blood samples from CM

patients, but downregulated in acral melanoma cells[22,30]. Moreover, miR-200c is known as a tumor suppressor that has been associated with the prognosis and metastasis of cutaneous melanoma[31]. The downregulation of miR-221 in plasma exosomes was consistent with other findings in blood samples. However, other findings in serum samples revealed that miR-221 was upregulated and significantly associated with patient survival[32,33]. Despite these promising biological associations, several critical research gaps have hindered the translation of EV-miRNAs into clinical reality. The current standard methods for EV separation and miRNA extraction are relatively ineffective and inefficient. Specifically, EV separation can result in substantial loss of vesicles, while miRNA extraction methods are complicated. Consequently, these technical hurdles significantly challenge the application of EV-miRNAs in clinical settings[34].

In this systematic review and meta-analysis, miR-16 was included in two studies. A study by Guo et al.[19] assessed the diagnostic performance of miR-16 as an individual miRNA, whereas Stark et al.[23] evaluated it in a miRNA combination panel, namely MELmiR-7. Although both studies reported dysregulation of miR-16, there were substantial differences in the direction of expression. Guo et al.[19] displayed a downregulation of miR-16, while Stark et al.[23] was upregulated.

Downregulation of miR-16 has been shown in melanoma cell lines compared with normal melanocytes. The expression of miR-16 is regulated by ETS1, a transcription factor that is consistently upregulated in melanoma. Elevated ETS1 binds to miR-16 and suppresses its tumor suppressive effects. Moreover, miR-16 inhibits melanoma progression and metastasis through modulation of the NF-$\kappa\beta$ pathway[35].Consistent dysregulation of miR-16 supports its potential as a focus for diagnostic research. However, further studies are needed to resolve the inconsistent direction of its expression in blood-based samples.

Heterogeneity was evaluated using the I^2 value in univariate analysis and the shape of SROC curve. The results of the I^2 statistics suggested substantial heterogeneity in the overall, individual miRNA, and miRNA combination only analyses. These results were further confirmed by visual inspection of the SROC curve, which displayed scattering of individual studies, especially in overall and miRNA combination SROC curves.

The potential source of heterogeneity was the presence of a threshold effect. The negative correlation (ρ) observed in this analysis indicated that heterogeneity between studies might occur as a result of threshold variation[36]. In addition, other potential contributors to heterogeneity include differences in cancer stages and assessment methods. For instance, a study by Armand-Labit et al.[17] exclusively included metastatic CM (stage III to IV), whereas the rest included all stages. With respect to miRNA assessment methods, variations have been reported in several miRNAs combination studies. A study by Stark et al.[23] and Van Laar et al.[24] applied independent scoring rather than using the expression value of each miRNA, while Leidinger et al.[22] used support vector machine-based classification to determine the probability of melanoma patients or healthy controls. These distinct analytical methods may explain the substantial heterogeneity observed.

Several limitations were encountered in this review. The number of included studies was relatively small; therefore, several further analyses including Egger's regression and Begg's correlation tests were not performed. In addition, substantial heterogeneity was observed, which likely arose from differences in diagnostic thresholds, assessment methods, and cancer stages across studies. Notably, conflicting expression profiles were observed for miR-16, which was reported to be downregulated by Guo et al.[19] but upregulated by Stark et al.[23] This discrepancy underscores a biological heterogeneity that complicates the clinical interpretation and standardization of these miRNAs as biomarkers. Finally, while their biological potential is promising, only one study on EV-miRNAs was eligible for analysis, hence their diagnostic performance could not be reliably assessed. The relatively inefficient EV separation and miRNA extraction may have hindered its translational applicability[34], which resulted in a less prevalent method among others.

Based on this systematic review and meta-analysis, blood-based circulating miRNAs displayed excellent diagnostic performance with AUC of 0.917 for overall studies. Notably, plasma samples and combination miRNAs showed superior diagnostic performance compared to other samples and individual miRNA. However, these findings should be interpreted in light of the predominance of case-control study designs, which are associated with a high risk of patient selection bias, as well as heterogeneity in the clinical stage across the included studies.

Methods

Protocol and registration

This study was carried out according to our protocol in PROSPERO (CRD420251125262) and Preferred Reporting Items for a Systematic Review and Meta-analysis (PRISMA) of Diagnostic Test Accuracy (DTA) Statement. The PRISMA-DTA checklist is presented in Supplementary Figure S1.

Eligibility criteria

Blood-based (whole blood, serum, and plasma) circulating miRNAs diagnostic accuracy studies comparing CM patients and nonmelanoma controls were the inclusion criteria. The exclusion criteria were studies with nonhuman patients and studies with insufficient data to construct 2×2 contingency tables.

Search strategy

Articles were searched in PubMed, EBSCO, and ProQuest using terms ("microRNAs" OR "circulating microRNAs" OR "miRNAs" OR "miRNA" OR "miR-*") and ("melanoma" OR "melanoma, cutaneous malignant" OR "cutaneous melanoma") and ("diagnostics") and ("expression"). These search terms are adapted for each database. Additional sources are obtained through citation searching. Only English-language articles published within the last 25 years (August 19, 2000-August 19, 2025) with accessible full texts were included. Full search strategies for each database are provided in Supplementary Table S2.

Study selection and data collection

The literature search and screening of studies were performed by two independent authors (H.P. and S.H.P.). Any discrepancies will be resolved through consensus with other authors (A.L.E. and F.M.S.). For each eligible study, sensitivity, specificity, and sample size data were extracted to construct a 2-by-2 contingency table comprising true positive (TP), false positive (FP), false negative (FN), and true negative (TN) data. In this review, the target condition was defined as a human patient with CM, an index test as any detection method for blood-based circulating miRNAs, and histopathological examination as a reference standard. For studies that reported multiple individual miRNA results within the same study, only the miRNA with the highest AUC was included in the meta-analysis.

Risk of bias and applicability

The risk of bias and applicability of the included studies were evaluated using QUADAS-2 tool. Evaluation of four key domains of the QUADAS-2 was performed by two independent authors (H.P. and S.H.P.) and any discrepancies were identified through consensus with other authors (A.L.E. and F.M.S.).

Data analysis

Studies reporting individual miRNA and miRNA combinations were analyzed separately. The diagnostic performance of both results was evaluated via bivariate models that paired sensitivity and specificity. Pooled diagnostic accuracy of sensitivity, specificity, and DOR were also performed using random effect meta-analysis with inverse variance methods. A modified Haldane-Anscombe correction was applied for studies with zero cells in their 2×2 contingency tables. Heterogeneity was assessed visually through the SROC curve and quantitively using the I^2 statistic and Cochran's Q tests. Between-study variance was estimated using restricted maximum-likelihood (REML) random-effects method. Interpretation of the I^2 statistics was performed according to the Cochrane Handbook for Systematic Reviews of Interventions. Significant heterogeneity was indicated by $P < 0.05$ for Cochran's Q test. Sources of heterogeneity were explored using bivariate meta-regression and publication bias was assessed using Deeks' funnel plot. The overall results of the selected studies were visualized using SROC curves. The performance of blood-based circulating individual miRNA and miRNA combinations for the diagnosis of CM was measured according to the AUC.

Data availability

The datasets used and/or analysed during the current study available from the corresponding author on reasonable request.

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Author contributions

H.P. and A.L.E. contributed equally as first authors to the publication. H.P., A.L.E., and F.M.S. were responsible

for study conception. H.P. and S.H.P. designed the study and independently screened the records, with A.L.E. and F.M.S. resolving any discrepancies. H.P. and S.H.P. performed the full data analysis. All the authors contributed to data interpretation, and manuscript preparation, and approved the final version for publication.

Declarations

Competing interests

The authors declare no competing interests.

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Correspondence and requests for materials should be addressed to F.M.S.

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